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Professional Summary

ECE undergraduate specializing in Semiconductor VLSI and Chip Design Verification. Hands-on experience with RTL design, RISC-V architecture, and building self-checking testbenches. Strong foundation in digital logic, computer architecture, and EDA tools (Yosys, OpenLane) to drive rigorous pre-silicon verification.

Education

Aug 2023 – May 2027 **B.Tech, Electronics and Communication Engineering**, *Vellore Institute of Technology*, Bhopal, *CGPA: 8.66/10*

Technical Skills

HDL & Verif. SystemVerilog, Verilog, cocotb, Testbench Development, Code Coverage
EDA Tools Yosys, OpenLane, Icarus Verilog, Verilator, LTspice, MATLAB
Digital Design RISC-V Architecture, Pipelining, Digital Logic, FSMs, Microcontrollers
Protocols UART, SPI, I2C, Wishbone Bus, PWM, GPIO, ESP32
Programming C, C++, Python, Embedded C, Bash/Linux Scripting

Projects

Oct 2026 – Nov 2026 **MiniSoC-32: RISC-V Based System-on-Chip (RTL to GDSII)**
Stack: SystemVerilog, Yosys, OpenLane, cocotb, SkyWater 130nm
- Designed a 32-bit SoC in SystemVerilog with a 5-stage pipelined RV32I core, Wishbone bus, 4KB ROM/RAM, and memory-mapped UART, SPI, GPIO and timer peripherals.
- Built a self-checking verification environment using cocotb/Icarus Verilog, achieving [X]% pass rate across [N] test cases with lint-clean RTL and [X]% code coverage.
- Synthesized design from RTL to GDSII using Yosys and OpenLane, closing timing at [50/100] MHz with [X] mm^2 area, [X]% utilization and [X] mW power.

Jul 2026 **AI-Powered RF Spectrum Intelligence Platform**
Stack: Python, DSP, TensorFlow, CNN
- Architected a DSP pipeline (AWGN noise, low-pass filtering, FFT) for automatic RF signal generation and BPSK/QPSK/OFDM analysis.
- Trained a CNN on spectrogram features achieving over 90% validation accuracy, deployed via a Streamlit dashboard.

Feb 2025 – Apr 2025 **QUENCH - IoT Smart Water Dispenser**
Stack: Embedded C/C++, ESP32, Sensors
- Engineered board design for a contactless IoT dispensing system, integrating an ESP32 and managing power distribution.
- Achieved response time under 2s and 95% flow measurement accuracy through precise sensor calibration.

Experience

Jun 2026 – Aug 2026 **Embedded System Design Intern**, *Maven Silicon*
- Completed a structured curriculum covering digital design fundamentals, core microcontroller architecture, and GPIO.
- Executed system-level unit and integration tests on sensor inputs and digital control logic.

May 2026 – Jul 2026 **AI/ML Intern**, *MPOnline Limited*
- Developed data preprocessing pipelines and trained deep learning models using Python and TensorFlow.
- Deployed an end-to-end AI capstone project utilizing Git-based version control and MLOps practices.

Certifications

VLSI Core Fundamentals of Digital Design for VLSI Chip Design (L&T EduTech / Coursera)
Certifications NPTEL Certified in IoT (88%); Applied Machine Learning in Python (88.6%)